

Final Project Proposal

Format and Guidelines

CSS142P Modelling and Simulation | School of Information Technology, Mapua University

Proposal task: Name the system you will simulate, draw its boundary, choose the model type, and commit to the performance measures your simulation will report.

Required result: An approved proposal. No group may begin building the final project model until its proposal is approved by the instructor.

1. Required Structure

Use these headings, in this order, with these exact names.

Part	Heading	What it must contain
0	Cover Block	Working project title, group number, full names of all members, date, one-line statement of the system in plain language.
1	Problem Statement and Decision	The system, who runs it, what goes wrong with it now, and the decision a simulation would inform. Name the decision explicitly.
2	System Boundary	Entities, resources, the arrival process, the start and stop conditions of a run, and an explicit list of what is left outside the model.
3	Model Type and Justification	Discrete-event, Monte Carlo, or continuous, and why that type fits the system. Name the state variable or the event set.
4	Performance Measures	Three to five measures, each with a unit and a one-line statement of how it will be computed from the run.
5	Inputs and Tools	Every input the model needs, where each value will come from, and the tools you will use.

Part 2: what the boundary must state

- **Entities:** what moves through the system, and one concrete example.

- **Resources:** what entities compete for, and how many of each.
- **Arrivals:** how entities enter, and over what period.
- **Run boundary:** when a run starts, when it stops, and whether there is a warm-up.
- **Out of scope:** at least three things you are deliberately not modelling.

Part 4: what counts as a performance measure

A performance measure is a number your program prints at the end of a run. It has a unit, and someone running the real system would act differently if it changed. At least one measure must be a time, and at least one must be a count or a utilization.

Measure	Unit	How it will be computed
Average waiting time in queue	minutes	Mean over all entities of service start time minus arrival time.
Server utilization	fraction of time	Total busy time divided by total simulated time.
Maximum queue length	entities	Running maximum of the queue size, sampled at every event.

The three rows above are the level of detail expected. Do not submit "efficiency", "performance", or "customer satisfaction": none of these is a number a program can print.

2. Proposal Template

Copy this template into your own document and fill it in. Keep the row labels.

Field	Your entry
Working title	
Group number	
Members (full names)	
Date	
The system in one line	
Part 1: problem and decision	
Part 2: entities	
Part 2: resources	

Field	Your entry
Part 2: arrivals	
Part 2: run start and stop	
Part 2: out of scope	
Part 3: model type	
Part 3: why this type fits	
Part 4: measures, units, computation	
Part 5: inputs and where each value comes from	
Part 5: tools	

3. Worked Example

An approved-quality proposal, filled in. This is the level of detail expected in each field.

Field	Entry
Working title	Walk-in queue at a campus health clinic
The system in one line	Students arrive at the clinic without appointments, wait for one of two nurses, and leave after being seen.
Part 1: problem and decision	The clinic runs two nurses from 08:00 to 17:00. Students report waits of over an hour around midday, and the clinic head must decide whether a third nurse from 11:00 to 14:00 is worth the cost. The decision is: add a midday nurse, or leave staffing as it is.
Part 2: entities	Walk-in students. One student is one entity.
Part 2: resources	Nurses, two available, each serving one student at a time.
Part 2: arrivals	Students arrive one at a time through a nine-hour day, more heavily between 11:00 and 14:00.
Part 2: run start and stop	A run is one clinic day, 08:00 to 17:00. Students already in the queue at 17:00 are served before the run ends. No warm-up: the clinic starts empty each day.

Field	Entry
Part 2: out of scope	Referrals to the hospital, medicine stock, appointment bookings, nurse breaks.
Part 3: model type	Discrete-event simulation.
Part 3: why this type fits	The clinic state changes only at arrivals and at service completions, and the quantity of interest is queueing behavior rather than a continuously varying level. The event set is arrival, service start, and service completion.
Part 4: measures, units, computation	Average waiting time (minutes): mean of service start minus arrival over all students. Nurse utilization (fraction): busy nurse-time divided by nurse-time available. Maximum queue length (students): running maximum of the waiting line. Students seen per day (count): number of service completions per run.
Part 5: inputs and where each value comes from	Arrival rate per hour: counted from the clinic logbook for one week. Service time: timed for twenty consultations, fitted to an exponential distribution. Nurse count: clinic policy, known exactly.
Part 5: tools	Python 3.11, SimPy, NumPy. Thirty independent replications per configuration, fixed random seeds recorded.

4. Scope Guardrails

Most rejected proposals are rejected for size, in one direction or the other.

Too large

- A whole airport, hospital, or city. Model one counter, one ward, one intersection.
- A system whose inputs you cannot observe or defend. If you cannot say where an arrival rate comes from, you cannot use it.
- More than one decision. One decision per project.

Too small

- A system with no queue, no contention, and no randomness. If every entity is served immediately, there is nothing to simulate.
- A problem with a closed-form answer you could compute by hand in five minutes.

5. How the Proposal Is Judged

The instructor returns one of three decisions within one week of the deadline.

Decision	Meaning
Approved	Begin building. The proposal is now the specification your final report is checked against.
Approved with conditions	Begin building, but apply the stated conditions. No resubmission required.
Not approved	Revise and resubmit within three days. The most common cause is a boundary that was never drawn, or measures that are not numbers.